

LESSON 03 / 第三課

Pruning as a Physiological Dose: From Crown Removal to the Cut Face

作為生理劑量的修剪：從樹冠移除到切面

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Pruning is the controlled removal of living and dead tissue. It is not a style. At crown scale the live dose is leaf area and stored carbon taken out of the tree. At branch scale the relevant geometry is aspect ratio and the collar. At the cut face the event is a CODIT wound whose diameter sets the carbon and hydraulic bill. Timing is a phenological and climatic envelope, not a sacred month. Gilman's aspect-ratio work, ANSI A300 Part 1, and the experimental pruning-dose literature converge on one result: large live cuts are expensive, co-dominant unions with ratio approaching 1 are structurally expensive, and no cut reprints lost section.

修剪是受控移除活組織與死組織。它不是一種風格。樹冠尺度上，活劑量是從樹上取走的葉面積與貯存碳。枝尺度上，相關幾何是直徑比與枝領。切面上，事件是一次 CODIT 傷口，其直徑設定碳與水力帳單。時間是物候與氣候包絡，不是神聖月份。Gilman 的直徑比工作、ANSI A300 第 1 部分，以及實驗性修剪劑量文獻收斂於一項結果：大活切面昂貴，直徑比趨近 1 的共優聯合在結構上昂貴，任何切面都不重印已失截面。



Figure 1. Nested scales from live dose at the crown to the ledger after the cut.

圖 1. 由樹冠活劑量到切後賬戶的嵌套尺度。

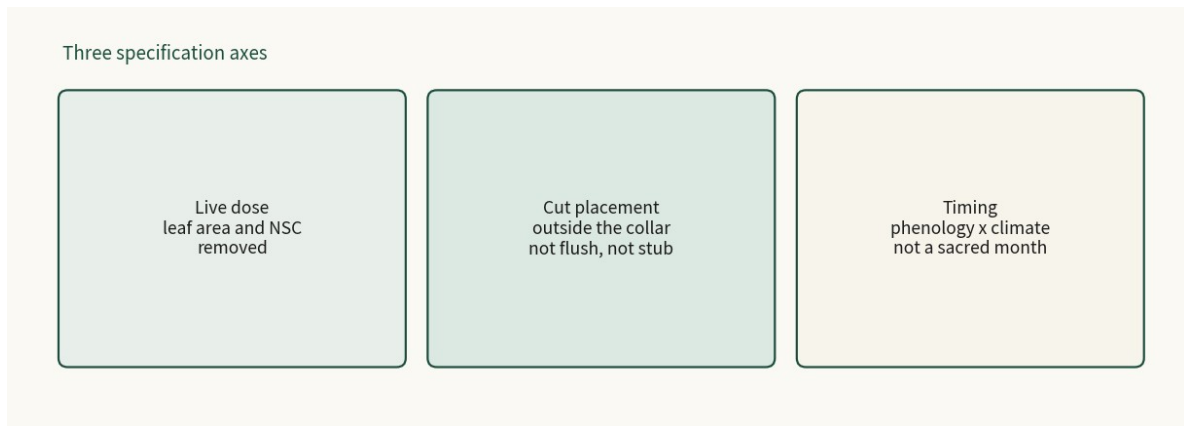


Figure 2. Three specification axes: live dose, cut placement, timing.

圖2. 三條規格軸：活劑量、切面位置、時間。

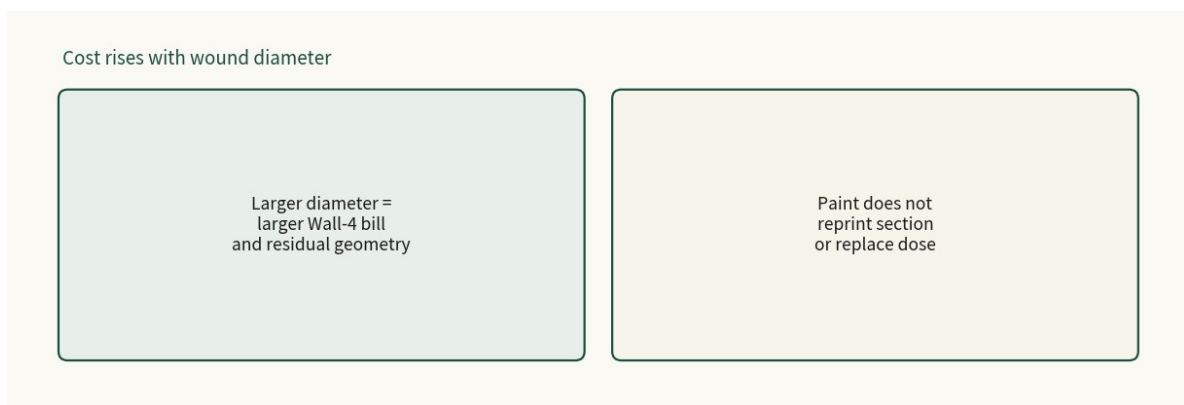


Figure 3. Compartmentalization cost rises with wound diameter.

圖3. 區隔化成本隨傷口直徑上升。

1. Scope

1. 範圍

The object is published pruning physiology and specification, from the fraction of live crown removed to the anatomy of the cut face. Landscape fashion is out of scope except where it has been tested as a dose.

對象是已發表的修剪生理與規格：由移除的活樹冠比例，到切面解剖。園景時尚不在範圍內，除非它已作為劑量被檢驗。

2. Live dose

2. 活劑量

Live pruning removes photosynthetic surface and a fraction of stored non-structural carbohydrate. Fini, Ferrini and colleagues, and the broader dose literature, treat the percentage of live crown removed as a physiological event with growth, hydraulic and defence consequences, not as a visual preference [1,2]. Severe live reduction can thin the carbon budget that pays for Wall 4 at the cut and for new absorbing roots in the same season. Deadwood removal is a different account: it does not spend the same live-leaf and NSC pool. Mixing the two percentages in a single ‘percent pruned’ number hides the dose that actually taxes the tree.

活修剪移除光合表面與一部分貯存的非結構性碳水化合物。Fini、Ferrini 及同事以及更廣的劑量文獻，把移除的活樹冠百分比當成具有生長、水力與防禦後果的生理事件，而不是視覺偏好 [1,2]。

重度活削減可抽薄同一季節支付切面第 4 壁與新吸收根的碳預算。清除枯木是另一本賬：它不花同一活葉與 NSC 庫。把兩個百分比混進單一「修剪百分比」，會藏起真正抽稅的那劑量。

3. Branch geometry and the cut

3. 枝幾何與切面

Gilman's aspect-ratio work showed that co-dominant unions with diameter ratio approaching 1 are structurally expensive relative to unions with a clear subordinate [3]. That result is independent of how clean the cut looks after the fact. The collar is the anatomical landmark for a reduction or removal cut; a flush cut removes the collar's defensive tissues, and a stub leaves a column the tree must wall off twice [4,5]. ANSI A300 Part 1 writes those landmarks as specification language: cut type, dose and objectives must be stated [5]. A specification that says only 'crown reduce' has not yet specified a dose.

Gilman 的直徑比工作顯示，直徑比趨近 1 的共優聯合，相對有明確從屬枝的聯合，在結構上更昂貴 [3]。該結果獨立於事後切面看起來多乾淨。枝領是縮減或移除切的解剖地標；平切去掉枝領的防禦組織，殘樁留下樹木必須封兩次的柱 [4,5]。ANSI A300 第 1 部分把這些地標寫成規格語言：切面類型、劑量與目標必須寫明 [5]。只寫「樹冠縮減」的規格尚未指定劑量。

4. Timing and the cut face

4. 時間與切面

The published constraint is phenology interacting with climate, not a universal month [1,6]. Cutting immediately before a flush spends reserves the flush also needs; cutting into drought or heat spends water and carbon already under demand. Tropical and Mediterranean calendars do not copy temperate dormancy rules. Every live cut is a wound whose diameter sets the area of Wall-4 work [4]. 'Cleaning the cut' with paint is not supported as a general improvement of compartmentalization in the modern experimental literature [5].

已發表約束是物候與氣候的相互作用，不是普遍月份 [1,6]。在抽梢直前切，花掉抽梢也需要的貯備；切入乾旱或高溫，花掉已在需求下的水與碳。熱帶與地中海日曆不複製溫帶休眠規則。每一次活切都是傷口，其直徑設定第 4 壁工作的面積 [4]。以漆「清潔切面」，在現代實驗文獻中並不作為區隔化的普遍改進而成立 [5]。

5. Limits

5. 限度

Most dose experiments are on temperate amenity taxa of modest size. Veteran retrenchment, palm pruning and tropical street trees have thinner experimental dose-response curves. Transfer of a percentage from a 20-year-old maple to a 200-year-old oak is not a published result.

多數劑量實驗在體型不大的溫帶園景分類群上。老齡回縮、棕櫚修剪與熱帶行道樹的實驗劑量-反應曲線較薄。把百分比從 20 年生楓轉移到 200 年生櫟，不是已發表結果。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Live pruning dose is first a:

1. 活修剪劑量首先是：

A. Style.

A. 風格。

B. Removal of leaf area and stored carbon.

B. 移除葉面積與貯存碳。

C. A t/R law.

C. t/R 定律。

D. Soil texture.

D. 土壤質地。

2. Deadwood removal and live-crown removal:

2. 枯木清除與活樹冠移除：

A. Are the same carbon account.

A. 是同一碳賬。

B. Are different accounts.

B. 是不同賬戶。

C. Both raise P50.

C. 都提高 P50。

D. Both are Wall 4.

D. 都是第 4 壁。

3. Gilman aspect ratio near 1 means:

3. Gilman 直徑比接近 1 意味：

A. A cheap union.

A. 便宜的聯合。

B. A structurally expensive co-dominant union.

B. 結構上昂貴的共優聯合。

C. A mycorrhizal rule.

C. 菌根規則。

D. A flash-density map.

D. 閃擊密度圖。

4. A flush cut is costly because it:

4. 平切昂貴是因為它：

A. Preserves the collar.

A. 保留枝領。

B. Removes collar defensive tissues.

B. 去掉枝領防禦組織。

C. Increases soil oxygen.

C. 增加土壤氧。

D. Sets 10-20-30.

D. 設定 10-20-30。

5. ANSI A300 Part 1 requires that dose and cut type be:

5. ANSI A300 第 1 部分要求劑量與切面類型被：

A. Inferred from a photograph.

- A. 從照片推論。
- B. Stated in the specification.
- B. 寫在規格裡。
- C. Replaced by paint.
- C. 用漆取代。
- D. Equal to canopy percent.
- D. 等於樹冠百分比。

6. Timing is specified as:

6. 時間被指定為：

- A. A universal sacred month.
- A. 普遍神聖月份。
- B. A phenological and climatic envelope.
- B. 物候與氣候包絡。
- C. Only i-Tree.
- C. 只有 i-Tree。
- D. A register listing.
- D. 名錄登錄。

7. Wound dressings as a general aid to CODIT are:

7. 傷口敷料作為 CODIT 的普遍助益：

- A. Strongly supported.
- A. 受到強力支持。
- B. Not supported as a general improvement.
- B. 不作為普遍改進而成立。
- C. Required by hydraulics.
- C. 水力所要求。
- D. Equal to P88.
- D. 等於 P88。

8. Larger cut diameter:

8. 較大切面直徑：

- A. Is free.
- A. 免費。
- B. Raises the CODIT bill and residual geometric risk.
- B. 提高 CODIT 帳單與剩餘幾何風險。
- C. Restores section.
- C. 恢復截面。
- D. Cools the city by 2 C.
- D. 使城市降溫 2 度。

9. A specification that says only ‘crown reduce’:

9. 只寫「樹冠縮減」的規格：

- A. Has specified a dose.
- A. 已指定劑量。
- B. Has not yet specified a dose.
- B. 尚未指定劑量。

C. Is a vulnerability curve.

C. 是脆弱性曲線。

D. Is Wall 1.

D. 是第 1 壁。

10. Transfer of a percentage dose from a young maple to a veteran oak is:

10. 把百分比劑量從幼楓轉到老齡櫟：

A. A published universal.

A. 已發表的普遍法則。

B. Not a published result.

B. 不是已發表結果。

C. Required by Santamour.

C. Santamour 所要求。

D. A palm rule.

D. 棕櫚規則。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

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[2] Pinkard E.A., Beadle C.L. physiological responses to green pruning.

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[5] ANSI A300 Part 1 (2023). Pruning.

[6] Lonsdale D. timing and decay-risk discussions in amenity-tree pruning.

[7] Smiley E.T., Kane B. pruning and structural failure literature.

[8] Clark J.R., Matheny N. pruning objectives and specification language.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。